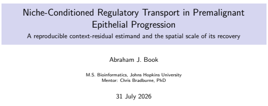
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

A reproducible context-residual estimand and the spatial scale of its recovery

Abraham J. Book

M.S. Bioinformatics, Johns Hopkins University
Mentor: Chris Bradburne, PhD

31 July 2026



Thank you all for being here. My name is Abraham Book, and this is work done with Chris Bradburne. I want to open with the question rather than the method, because the method only exists to answer it. A healthy pancreas contains hundreds of precancerous lesions, and almost none of them will ever become cancer. I want to know what holds them back — and specifically, whether what holds them back is inside the epithelial cell, or in the tissue around it.

The answer I am going to give you is a bounded one. The tissue does carry information about progression. And I can tell you the spatial scale at which that information stops being recoverable from the data we currently have.

[that second sentence frames the negative as a finding, in the first minute]

The paradox that defines premalignancy

concept schematic

a lesion field —
many indolent, one progressing

Same mutations.

Same histology.

Different outcomes.

The paradox that defines premalignancy
concept schematic
a lesion field —
many indolent, one progressing
Same mutations.
Same histology.
Different outcomes.

A grossly normal adult pancreas contains hundreds of precancerous lesions. Almost all of them carry oncogenic KRAS mutations. Almost none of them will become cancer.
In the airway the picture is even sharper. Carcinoma in situ lesions that are microscopically indistinguishable from one another have been followed longitudinally — and roughly half progress to invasive disease, while the other half regress or simply persist.
So the driver mutations are present. The histology is identical. And the outcomes diverge.
That is why histopathologic stage, which is indispensable for classification, does not tell you which lesion within a stage is going to move. And it is why molecular profiling of the epithelium alone has not closed the gap either.

[Braxton 2024, Teixeira 2019 — say them, do not slide them] [if asked why not just sequence the epithelium:

Anderson 2026 separates LUAD precursors within one histologic class, and the worst-outcome group differs as much by immune context as by epithelial state]

Cell-intrinsic

the tissue is a bystander

Tissue-held

the tissue is a participant

**Holding a cell's own state and stage fixed,
does its tissue context change where it ends up?**

Two explanations
Cell-intrinsic
the tissue is a bystander
Tissue-held
the tissue is a participant
Holding a cell's own state and stage fixed,
does its tissue context change where it ends up?

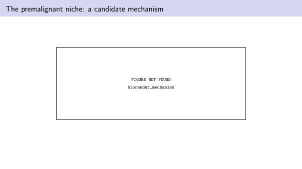
There are two broad explanations for that divergence.
One is cell-intrinsic: progression is set by the epithelial cell's own regulatory state, and the tissue is essentially a bystander.
The other is tissue-held: comparable epithelial cells progress differently because their surroundings differ. On this account the tissue is a participant, not a setting.
These make different predictions about one measurable quantity — the one on the slide. Holding a cell's own regulatory state and its stage fixed, does its tissue context change where it ends up?
Everything after this is the construction of that quantity and the attempt to falsify it.

[fifteen seconds — do not elaborate, the method is the next act]

The premalignant niche: a candidate mechanism

FIGURE NOT FOUND

biorender_mechanism



Before the method, the biological hypothesis in one picture. A premalignant epithelial lesion — AAH or AIS — sits inside a remodelling microenvironment: an IL1B-high macrophage niche, activated fibroblasts, and an immune compartment that is spatially organised rather than diffuse. The candidate mechanism is that signals from that niche — routed through receptors on the epithelial cell — redirect its regulatory program as it progresses, over and above what the cell's own state would do.

That is the "tissue-held" arrow made concrete. The rest of the talk asks whether that arrow carries measurable, reproducible information, and at what spatial scale.

[this is the one slide where you tell the biology story — everything after is measurement. Keep it to the niche -¿ receptor -¿ epithelial-program arrow.]

Profiling destroys the tissue.

No cell is ever observed twice.

So the effect cannot be measured. It has to be *estimated*.

Which means the design has to be built around falsification, not fit.

The measurement will not cooperate
Profiling destroys the tissue.
No cell is ever observed twice.
So the effect cannot be measured. It has to be estimated.
Which means the design has to be built around falsification, not fit.

There is an obstacle, and it determines the entire design.
Profiling destroys the tissue. So every human precursor cohort is a cross-sectional snapshot: each stage is represented by different cells, different lesions, different patients. No cell is ever observed twice.
That means the question “does this cell’s neighbourhood change its trajectory” is not directly observable at all. It has to be estimated, through a model of correspondence between unpaired populations.
And once you are estimating rather than measuring, two failure modes open up.
First, a niche can be statistically enriched at a stage without altering the trajectory of any cell in it. Enrichment and effect are different claims.
Second — and this is the one that matters — a model given neighbourhood features will fit better than one denied them whether or not the *identity* of the neighbourhood carries any information. More features, better fit.
So the design has to be built around falsification rather than fit.

[this slide buys credit for the whole talk — when the negative arrives in Act III they will already know falsification was designed in, not retrofitted] [the second failure mode motivates the shuffle control; call back to it later]

Three hypotheses, fixed before the data

H_1 A reproducible context residual exists.
H_2 That information is *receiver-local*.
H_3 The estimand transfers to a second epithelium.

They do not resolve in the same direction.
That is the contribution.

Three hypotheses, fixed before the data
H_1 A reproducible context residual exists.
H_2 That information is *receiver-local*.
H_3 The estimand transfers to a second epithelium.
They do not resolve in the same direction.
That is the contribution.

Three hypotheses were specified before the cohorts were analysed.
The first is that a reproducible context residual exists at all — that named regulatory programs carry a context effect whose sign holds up across held-out patients.
The second is that the information is receiver-local. Concretely: a model given each cell's own local neighbourhood should beat a model given a specimen average, should beat a model given no context, and — critically — should beat a model given context whose receiver correspondence has been destroyed but whose distribution is preserved. That last comparison is the one that separates real ecological information from added flexibility.
The third is that the estimand transfers: applied unchanged to a different precursor epithelium, it should return interpretable programs where the cohort supports them and an empty set where it does not.
Each has a null. Each null was run. They do not resolve in the same direction, and I would argue that is the contribution rather than a disappointment.

[formalised Ch2 §2.7.4, assessed Ch3 §hypothesis-assessment] [if asked whether this was really pre-registered —

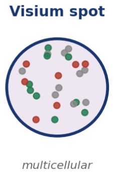
the identifiability screen rejected my own first design. That is slide 10.]

Act I

The estimand is identifiable, and it is not circular

Act I
The estimand is identifiable, and it is not circular

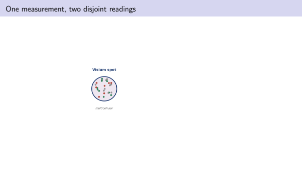
One measurement, two disjoint readings



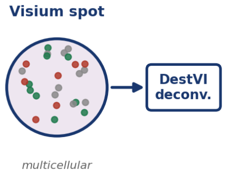
2026-07-31

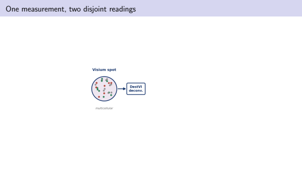
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└─One measurement, two disjoint readings

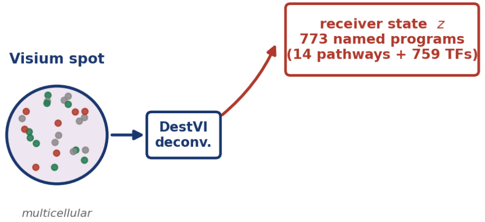


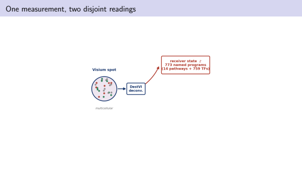
One measurement, two disjoint readings



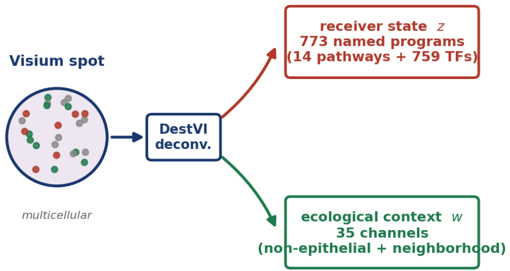


One measurement, two disjoint readings





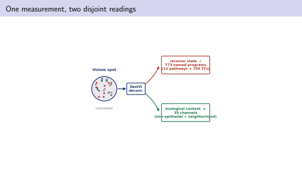
One measurement, two disjoint readings



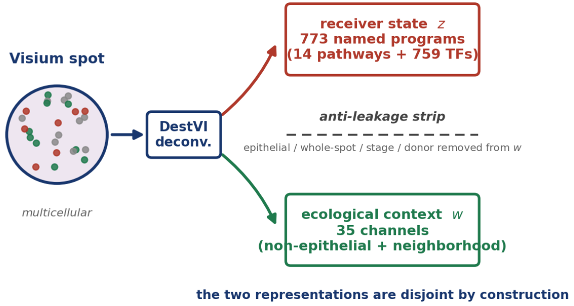
2026-07-31

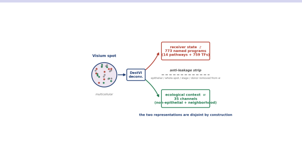
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└One measurement, two disjoint readings



One measurement, two disjoint readings

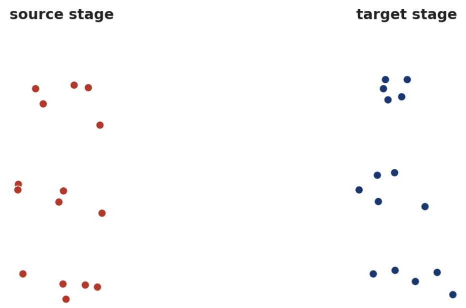




Here is how the two representations are built, and the important word is *disjoint*.
A spatial transcriptomic spot is a multicellular measurement. Deconvolution splits it in two.
From the epithelial side, I take the decoded epithelial expression and score it against curated pathway and transcription-factor priors. That gives the receiver state — 773 named programs: fourteen signalling pathways and seven hundred fifty-nine transcription factors. Not a latent space. Every coordinate has a name, which is what makes the output interpretable later.
From the other side, I take non-epithelial composition together with spatial neighbourhood features. That gives the ecological context, thirty-five channels.
And then an anti-leakage strip removes every epithelial, whole-spot, stage, and donor channel from the context vector before anything is fitted. The two representations do not share information by construction.

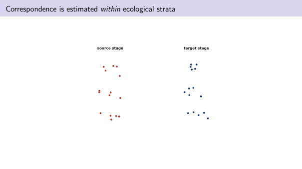
[be ready for "what exactly is a receiver?" — a Visium spot, epithelially decoded. The single-nucleus data is the deconvolution reference and the co-expression substrate, not a transport receiver. Know this cold.]

Correspondence is estimated *within* ecological strata

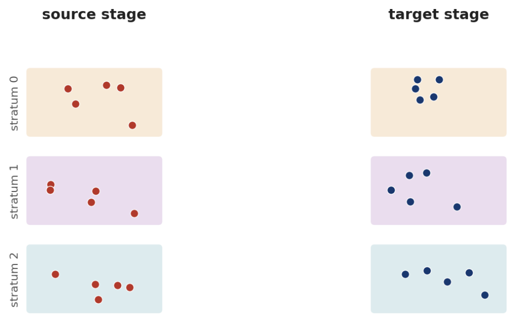


2026-07-31

Correspondence is estimated *within* ecological strata



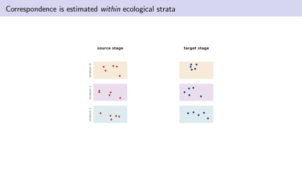
Correspondence is estimated *within* ecological strata



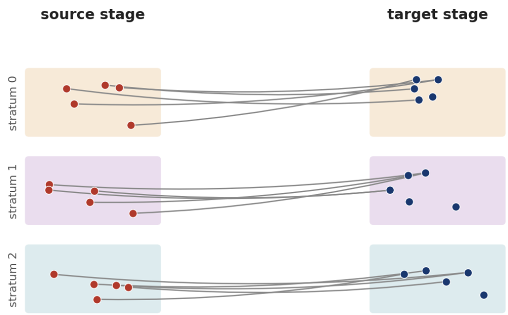
2026-07-31

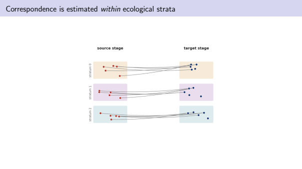
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

Correspondence is estimated *within* ecological strata

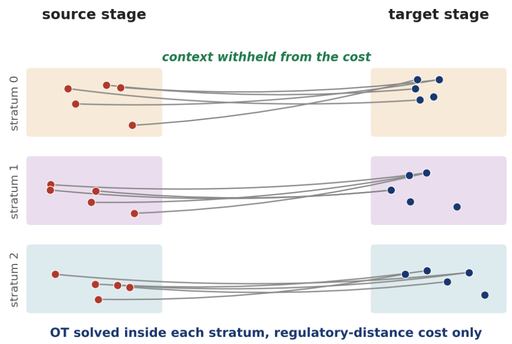


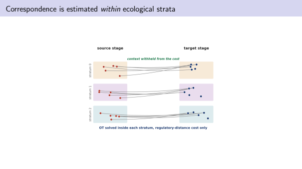
Correspondence is estimated *within* ecological strata





Correspondence is estimated *within* ecological strata





Now I need to relate source-stage cells to target-stage cells, and they are different cells from different patients.

Optimal transport gives a principled way to do that: treat the two populations as distributions and find the assignment of probability mass that minimises total cost.

But there is a trap. If I compute that coupling using context, then the coupling itself encodes the ecological differences I am trying to attribute to context. The effect gets absorbed into the correspondence used to estimate it, and the result is meaningless.

So the coupling is stratified. Broad composition defines coarse ecological strata, and transport is solved independently inside each one, using regulatory-state distance only. In the figure, the transport lines never cross a stratum boundary. Context is withheld from the cost entirely.

There is also a support gate: a stratum only enters if it holds at least ten receivers and at least two donors on both sides. So no correspondence is ever driven by an ecology seen in one stage or one patient.

[this design decision is the one most likely to be challenged — give the reason out loud, do not wait for the question]

The field is structured, and context enters through a waist

state z, τ

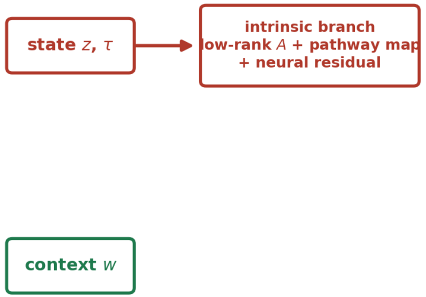
context w

The field is structured, and context enters through a waist

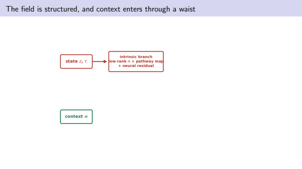
state z, τ

context w

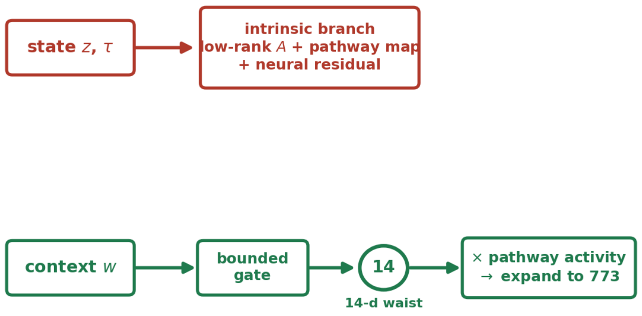
The field is structured, and context enters through a waist

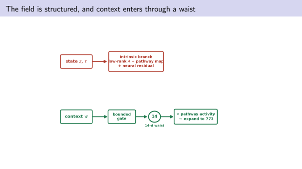


The field is structured, and context enters through a waist

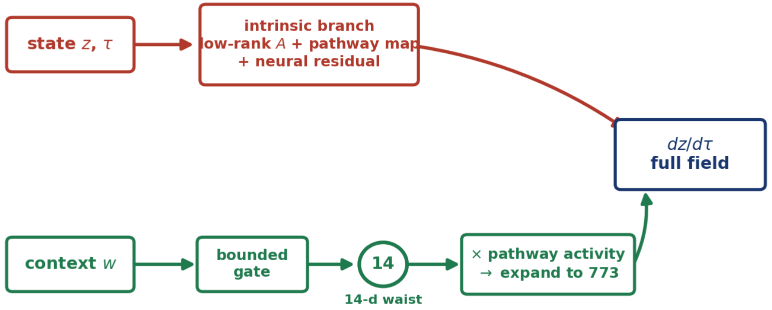


The field is structured, and context enters through a waist





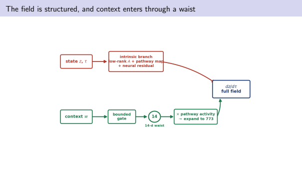
The field is structured, and context enters through a waist



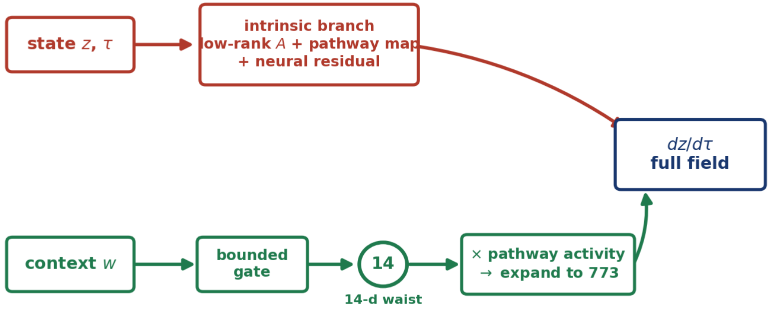
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

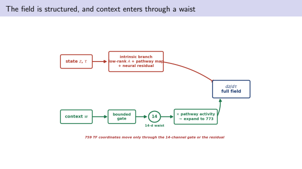
└The field is structured, and context enters through a waist



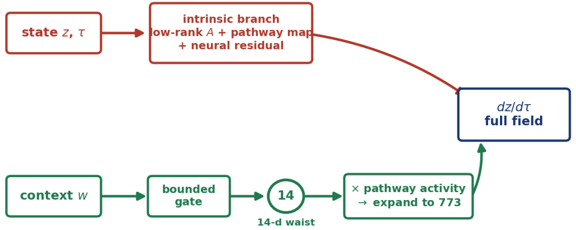
The field is structured, and context enters through a waist



759 TF coordinates move only through the 14-channel gate or the residual



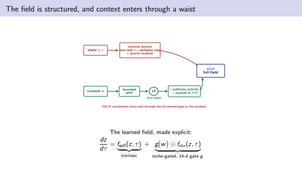
The field is structured, and context enters through a waist



759 TF coordinates move only through the 14-channel gate or the residual

The learned field, made explicit:

$$\frac{dz}{d\tau} = \underbrace{f_{\text{self}}(z,\tau)}_{\text{intrinsic}} + \underbrace{g(w)\odot f_{\text{ctx}}(z,\tau)}_{\text{niche-gated, 14-d gate } g}$$



This is the model. It is a differential equation on regulatory state, and it has two branches.

The intrinsic branch depends only on the cell's own state and its position along the stage edge. It has a rank-restricted linear operator, an explicit pathway-to-velocity mapping, and a neural residual.

The context branch takes the ecological vector, passes it through a bounded gate to produce one modulation score per pathway, multiplies that against the receiver's own pathway activities, and expands the result back into the full regulatory space.

And now the limitation, which I want to state before anyone asks me.

Context enters that structured field through only fourteen channels. The seven hundred fifty-nine transcription-factor coordinates are not gated directly — they move only insofar as they are driven through those fourteen, or through the neural residual. So when I show you transcription-factor level results later, those are downstream reflections of a low-dimensional gate, not independent per-factor evidence. The pathway-level results sit directly in the gated block and are the ones the architecture supports most strongly.

[anyone who reads the field equation finds this. Saying it first is worth more than being asked. Backup B2 has the full version.]

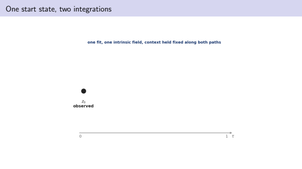
One start state, two integrations

one fit, one intrinsic field, context held fixed along both paths

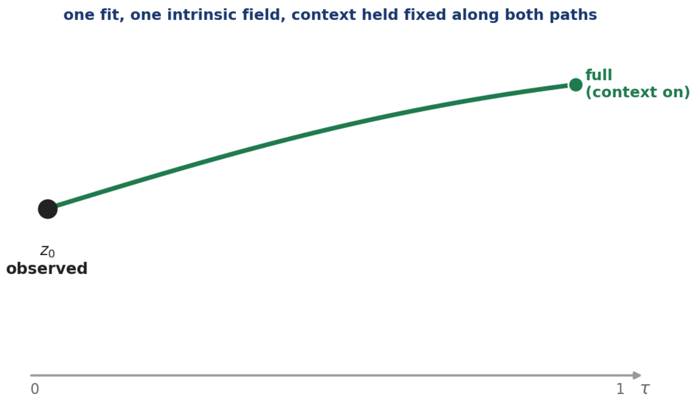
2026-07-31

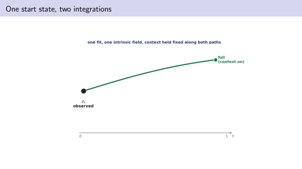
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└One start state, two integrations



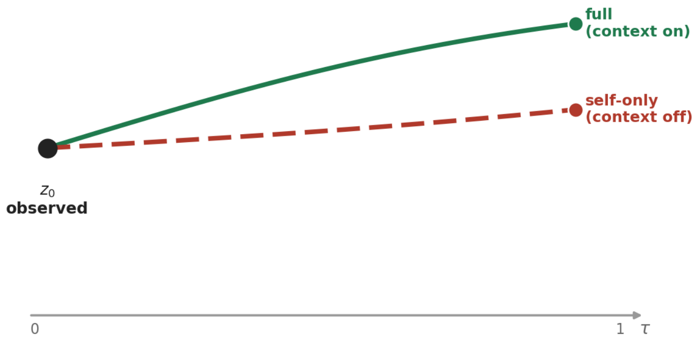
One start state, two integrations



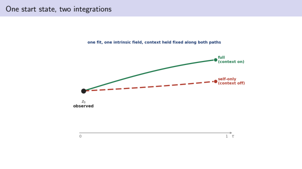


One start state, two integrations

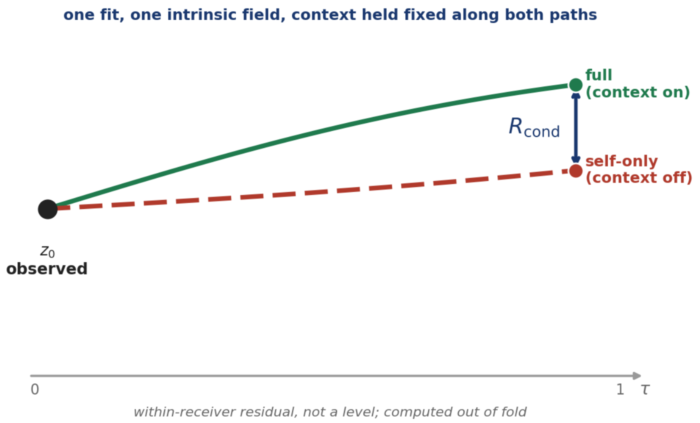
one fit, one intrinsic field, context held fixed along both paths



One start state, two integrations



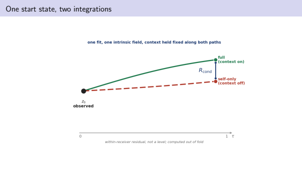
One start state, two integrations



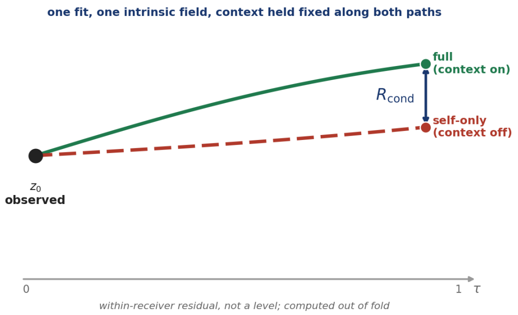
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

2026-07-31

└One start state, two integrations

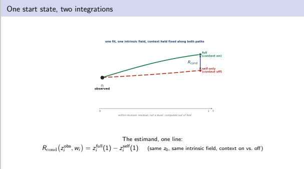


One start state, two integrations



The estimand, one line:

$$R_{\mathrm{cond}}\left(z_i^{\mathrm{obs}}, w_i\right) = z_i^{\mathrm{full}}(1) - z_i^{\mathrm{self}}(1) \quad \text{(same } z_0\text{, same intrinsic field, context on vs. off)}$$



And this is the estimand.

Take a cell in its observed regulatory state. Integrate the fitted field forward twice. Once with the context branch active. Once with it switched off. Same starting point, same fitted intrinsic dynamics, context held fixed along both paths.

The difference between the two endpoints is the conditional context residual.

Two things about that construction matter.

First, it is a *within-receiver* contrast, not a comparison between two separately fitted models. Both trajectories come out of one fit and share one intrinsic field. The gap is attributable to context rather than to two different models disagreeing.

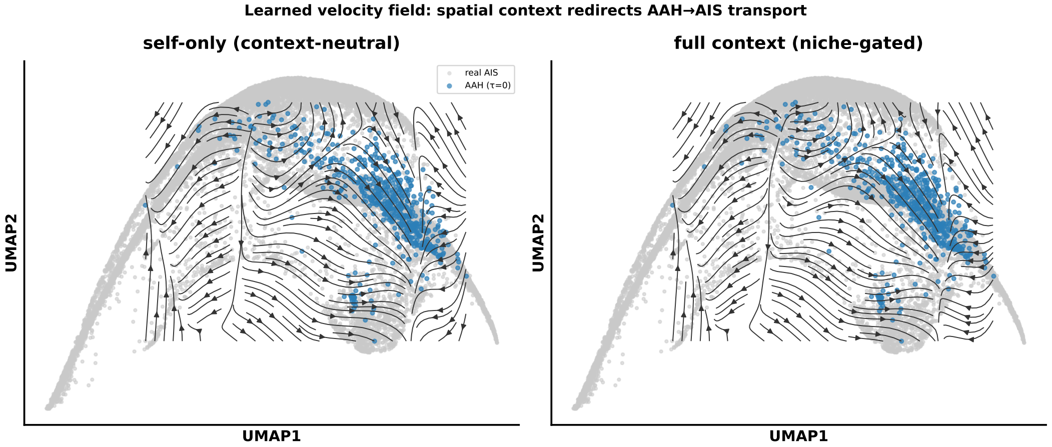
Second, it is a *residual*, not a level. It tells you how the surrounding tissue redirects a cell relative to where its own state would have taken it. It makes no claim about absolute abundance of anything. I will come back to that distinction, because it resolves a result that otherwise looks wrong.

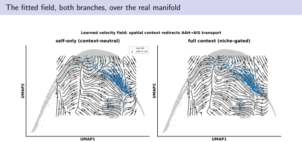
Everything I report is computed out of fold, on held-out patients only.

[say “out of fold, held-out patients” once here and you never defend it again] [plant “residual, not a level” —

you spend it twice: on VEGF, and on anyone who says a finding contradicts known biology]

The fitted field, both branches, over the real manifold



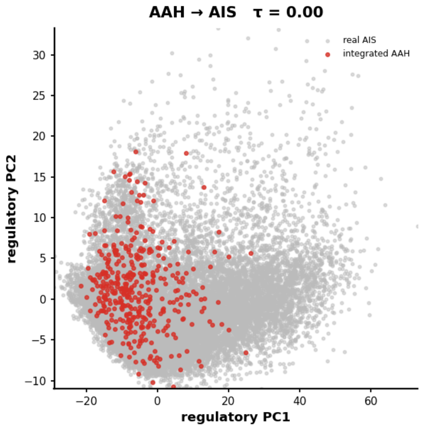


That was the schematic. This is the fitted field itself, drawn as streamlines over the real receiver manifold. Left is the self-only branch — the cell's own dynamics. Right is the full, niche-gated field. Blue are the AAH receivers at $\tau = 0$; grey is the real AIS target.

Two things to see. The flow is *directed* — it carries AAH cells toward the AIS region rather than relaxing to a point. And the two panels are nearly identical: the niche-gated field redirects transport only modestly. That is not a failure — it is the first visual statement of the result I will quantify later, that the context effect is real but small, and that its recoverable scale is the question of this thesis.

[do not over-claim the difference between the panels — the whole point is that it is subtle]

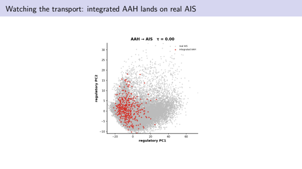
Watching the transport: integrated AAH lands on real AIS



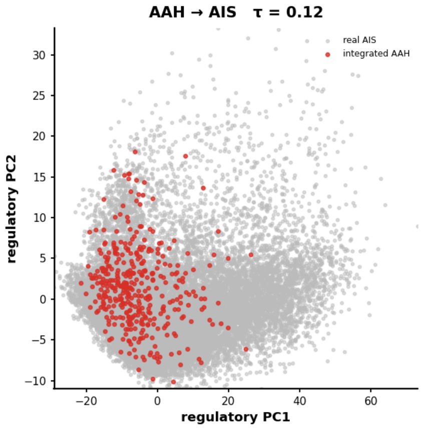
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└Watching the transport: integrated AAH lands on real AIS



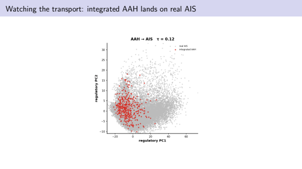
Watching the transport: integrated AAH lands on real AIS



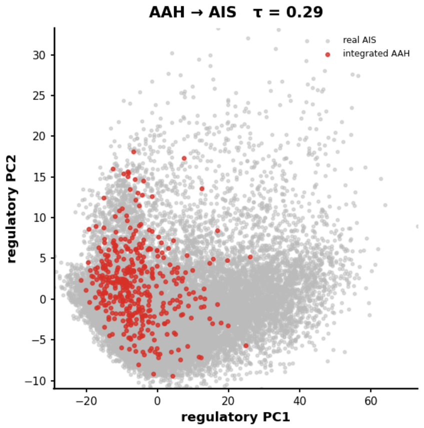
2026-07-31

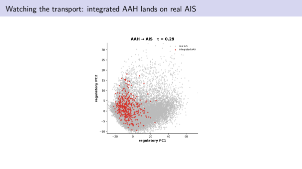
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└─Watching the transport: integrated AAH lands on real AIS

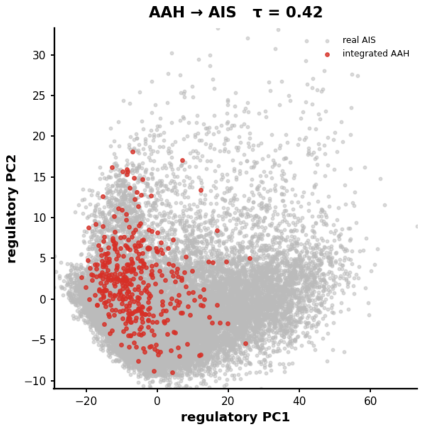


Watching the transport: integrated AAH lands on real AIS




Watching the transport: integrated AAH lands on real AIS

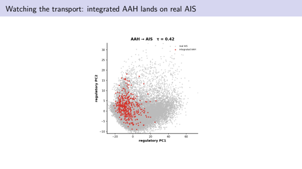
Watching the transport: integrated AAH lands on real AIS



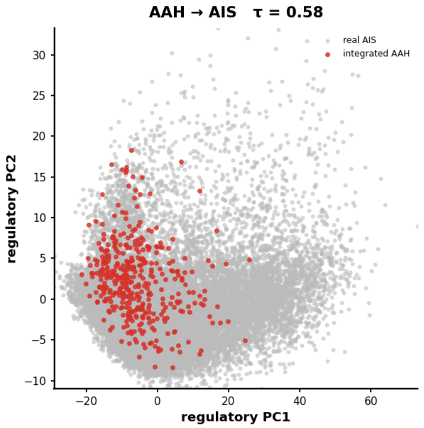
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└─Watching the transport: integrated AAH lands on real AIS



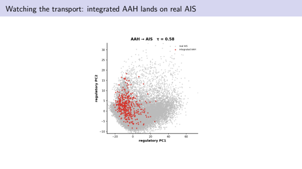
Watching the transport: integrated AAH lands on real AIS



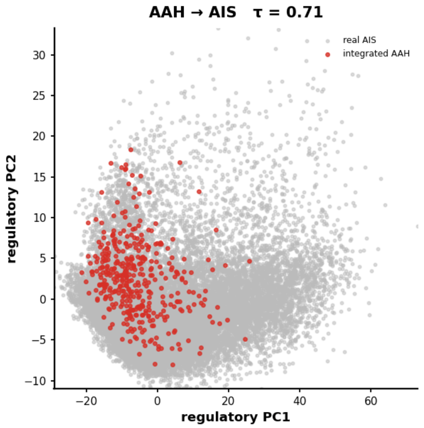
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└─Watching the transport: integrated AAH lands on real AIS



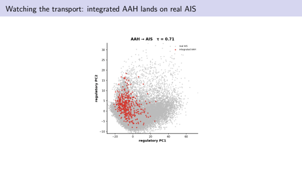
Watching the transport: integrated AAH lands on real AIS



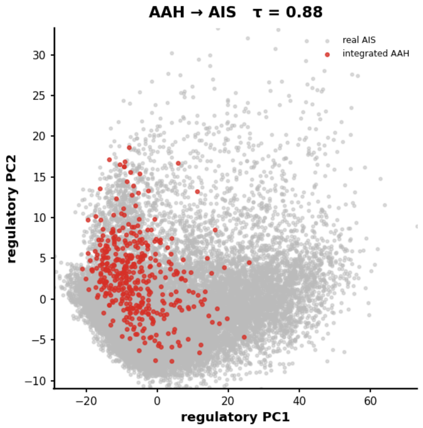
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└Watching the transport: integrated AAH lands on real AIS



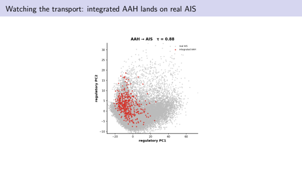
Watching the transport: integrated AAH lands on real AIS



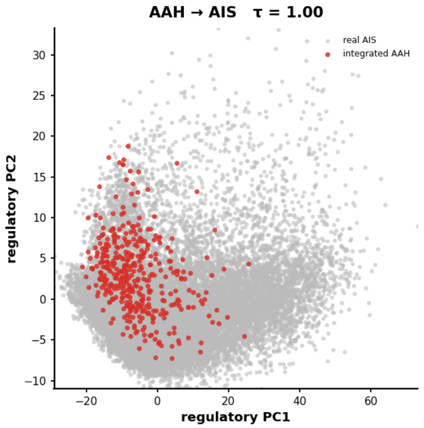
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

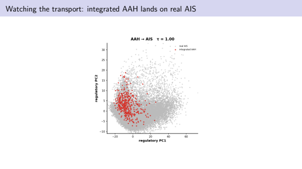
└─Watching the transport: integrated AAH lands on real AIS



Watching the transport: integrated AAH lands on real AIS



└Watching the transport: integrated AAH lands on real AIS



This is the fitted transport running on the primary edge, in the regulatory principal-component plane. Grey is the real AIS distribution — the target we never show the model. Red is the integrated AAH receivers, moving from their observed state at $\tau = 0$ toward $\tau = 1$.
Watch where they land. The red cloud settles into the body of the grey distribution. That endpoint agreement — integrated source onto held-out real target — is what the Sinkhorn endpoint metric scores, out of fold.

[if presenting from Adobe, slide 1 auto-plays the video; otherwise step 2–9 flips through the real frames. Either way it is the SAME model output, not a cartoon.] [one sentence only — "the model was never shown the grey."

Then move on.]

My first design failed its own screen

Three candidate estimands.

Two were rejected.

One of them was mine.

My first design failed its own screen

Three candidate estimands.

Two were rejected.

One of them was mine.

Before any of this touched real data, I built a synthetic benchmark with known ground truth and froze it — four acceptance thresholds, fixed in advance.

Three candidate coupling designs went through it.

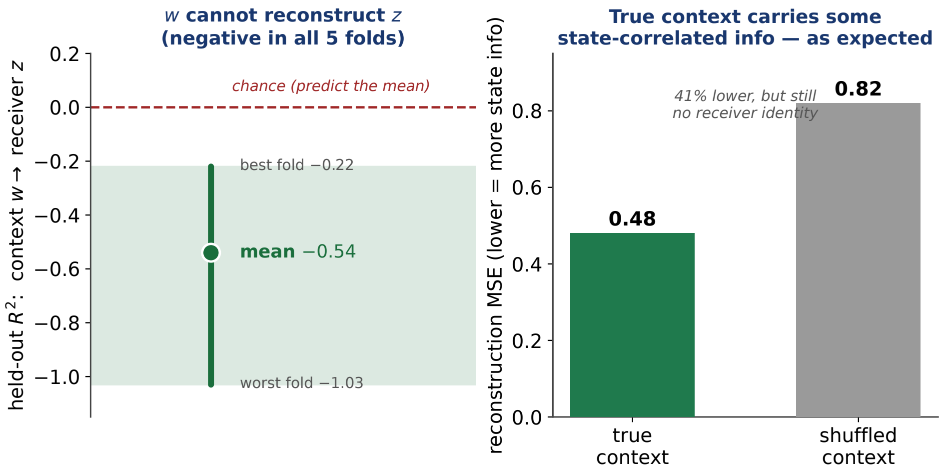
The progression-geometry coupling failed both tiers. The strict-state coupling — which was my first design, and the one I expected to use — passed the first tier and failed the second. Both were discarded.

The conditional-stratified coupling passed both, and it is what every result I am about to show you is built on.

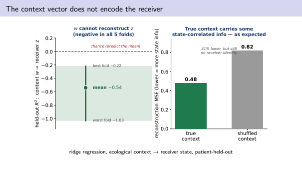
I want to be direct about why this slide is here. A falsification framework is only worth anything if it was capable of rejecting the thing you wanted to be true. Mine was, and it did. That is the evidence that the controls in this thesis are not post-hoc.

[do not skip this slide — it pre-empts “how do we know the controls weren’t chosen after the fact” entirely] [say the words: my pre-registered screen killed my first design and I changed it]

The context vector does not encode the receiver



ridge regression, ecological context → receiver state, patient-held-out



One more thing has to be true before any of this is interpretable.
If the context vector secretly contained a copy of the cell's own state, then any context effect would be self-referential — I would be measuring the model reading the cell back to itself. So I tested it rather than assuming it.
I fit a ridge regression from context to receiver state on training cells, and evaluated it on held-out cells. The held-out R-squared is negative in every one of the five folds, mean minus zero point five four. Context predicts receiver state worse than simply predicting the mean.
I will be precise about one detail. True context does reduce reconstruction error relative to a shuffled control — zero point four eight against zero point eight two. That is a real reduction, and it is expected, because tissue ecology and epithelial state are genuinely correlated in biology. But neither recovers receiver identity. The claim that matters is the negative R-squared.

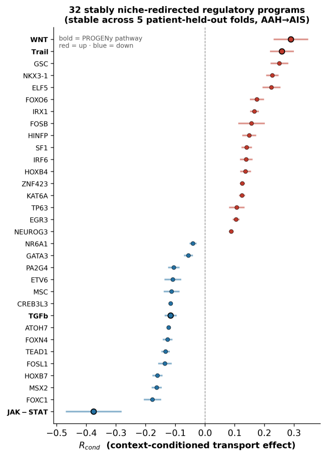
[do not call 0.48 vs 0.82 "marginal" — it is a 41% reduction, and overstating a control is how you lose it]

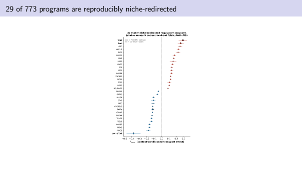
Act II

What the estimand finds

Act II
What the estimand finds

29 of 773 programs are reproducibly niche-redirected



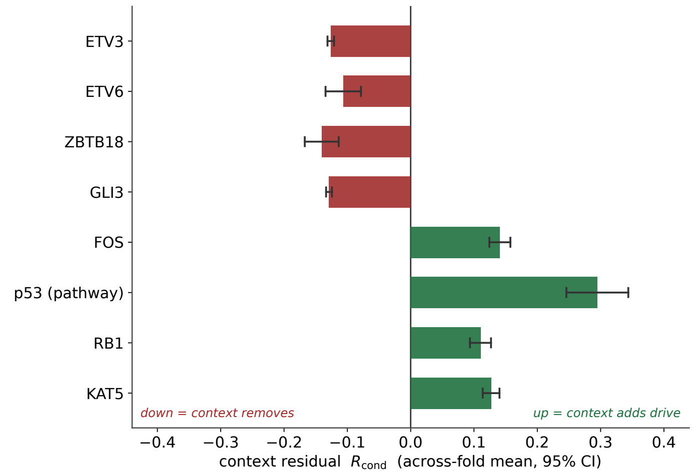


So: does a reproducible context residual exist? Yes.
Twenty-nine of the 773 programs on the preinvasive edge, and twenty-eight at the invasion boundary, carry a context residual whose sign holds in every one of five patient-held-out folds, with a donor-bootstrap interval excluding zero.
The leading programs also hold their sign in all ten leave-one-donor-out runs, so no single patient is carrying the set.
That is hypothesis one, met.

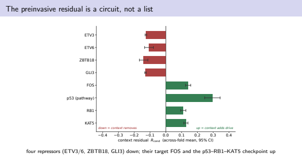
[expect: "29 out of 773 is 3.8% — isn't that just your false positive rate?" It is a fair question and you should say so.] [short answer: the criterion is not one test. It is a confidence interval excluding zero AND sign-consistency

across five disjoint patient folds. Under independence that compounds well below one percent. But the folds are correlated, so I do not claim an exact null rate — and the retrained-shuffle null, which is the correct test, is specified and not yet run. Backup B1. Do not bluff this.]

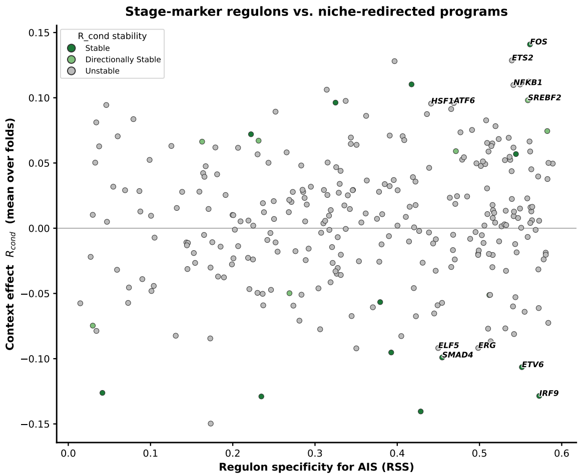
The preinvasive residual is a circuit, not a list



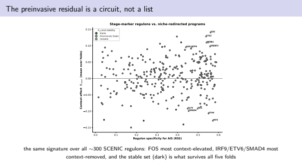
four repressors (ETV3/6, ZBTB18, GLI3) down; their target FOS and the p53–RB1–KAT5 checkpoint up



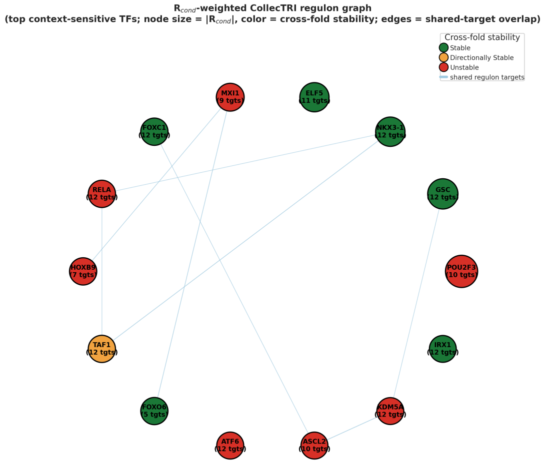
The preinvasive residual is a circuit, not a list



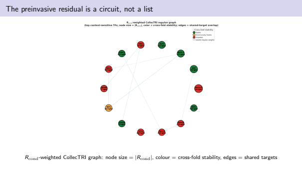
the same signature over all ∼300 SCENIC regulons: FOS most context-elevated, IRF9/ETV6/SMAD4 most context-removed, and the stable set (dark) is what survives all five folds



The preinvasive residual is a circuit, not a list



R_{cond}-weighted CollecTRI graph: node size = $|R_{\mathrm{cond}}|$, colour = cross-fold stability, edges = shared targets



This is the slide where I think the method earns its keep, so I want to spend a moment on it.
Read as a list, the stable set looks like a mixture. Read as a circuit, it resolves.
Four of the programs that lose context weight are transcriptional *repressors*. ETV3 and ETV6 are ETS-family repressors of immediate-early genes. ZBTB18 represses mesenchymal programs. GLI3, in its processed form, is the repressive output of Hedgehog signalling.
Four independent repressors move down. And FOS — the immediate-early factor they collectively restrain — moves up.
So this is not "AP-1 activity increases." It is de-repression: the brakes and their output are observed moving in opposite directions within the same estimate. That is a stronger statement, and it is only available because the readout is signed, per-program, and named.

[advance] And it is not a cherry-picked handful. Plotted over all three hundred SCENIC regulons against

their stage specificity, FOS is the single most context-elevated program and IRF9, ETV6 and SMAD4 sit at the context-removed extreme — the darkened points are the ones stable across all five folds, so the circuit I just described is the tail of a genome-wide distribution, not a list I selected.
At the same time, the p53, RB1 and KAT5 checkpoint axis is context-*elevated*. KAT5 is worth naming: it is the acetyltransferase that acts on p53. A cofactor and its pathway both coming out stable across

Same configuration. Different organ. Different method.

This work human LUAD precursors context-residual transport	**Reyes et al., Cell 2026** mouse PDAC, spontaneous p53 loss single-cell + spatial lineage
p53 ↑ RB1 ↑ SMAD4 ↓, with WNT / EGFR / AP-1 ↑	p53, CDKN2A, SMAD4 **co-activated** with oncogenic programs in progenitor-like cells

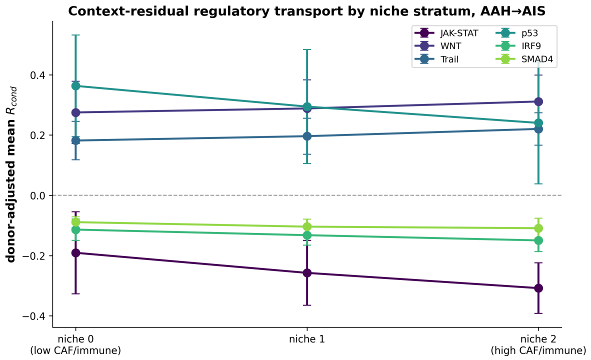
Including the same arm — SMAD4 — giving way.

Same configuration. Different organ. Different method.
Including the same arm — SMAD4 — giving way.

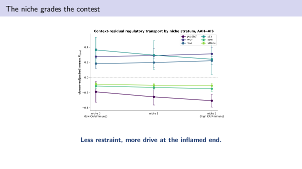
That configuration has been seen before, and not by me.
In mouse models of pancreatic carcinogenesis that capture spontaneous p53 loss, oncogenic and tumour-suppressive programs are found *co-activated* in a discrete progenitor-like population right at the benign-to-malignant transition — specifically the programs controlled by p53, CDKN2A, and SMAD4.
That is the same configuration I recover in human lung precursors, by a completely different route. And the correspondence extends to the detail: SMAD4 is the suppressive arm that gives way in their system, and SMAD4 is the suppressive arm that loses context weight in mine.
Two species. Two epithelia. Two unrelated inference strategies. Same result.
That kind of convergence is worth more than either observation alone, and it suggests the contested state is a general property of the precursor-to-invasion boundary rather than something specific to my cohort or my estimator.

[your strongest single slide — do not rush it] [Reyes also supplies causal direction: p53 suppression enables progenitor expansion, EMT reprogramming, and immune-privileged niche formation. That sets up the next slide.]

The niche grades the contest



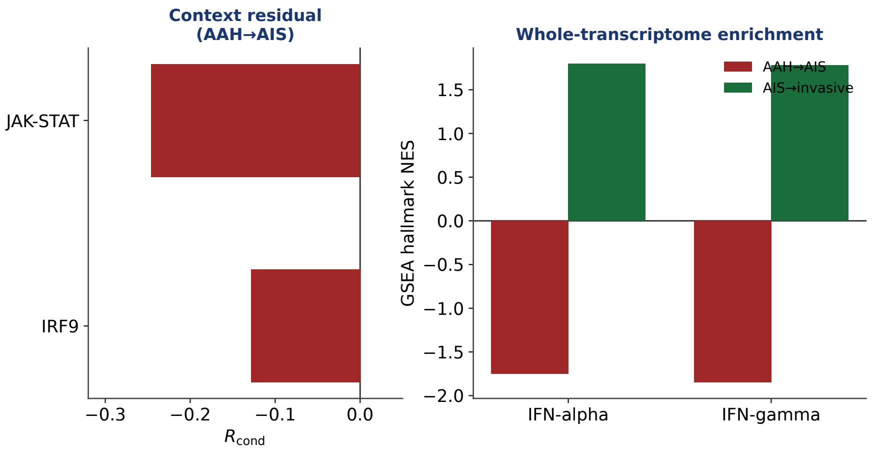
Less restraint, more drive at the inflamed end.



And the niche does not just add an effect — it grades it.
Within a single histologic class, I stratified receivers by how CAF- and immune-dense their surroundings are, and looked at how the residuals order across those strata.
Moving toward the inflamed, stroma-rich end: p53 restraint *falls*, from plus zero point three six three to plus zero point two four zero. WNT rises. TRAIL rises. All three significant against a within-donor permutation null at about five times ten to the minus four.
So the same lesion class shows less restraint and more drive where the tissue is more inflamed. That is the direction a permissive microenvironment would predict, and it is the direction the pancreatic model I just showed you predicts causally.
It is graded rather than switch-like, and the magnitudes are modest. I would not describe this as a niche licensing progression. I would describe it as the niche tilting a balance.

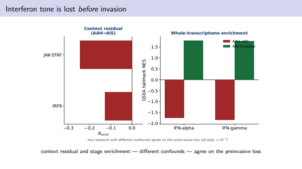
[state the permutation null before anyone asks — the strata also define the transport coupling, and that is a fair circularity concern. Permuting within donor preserves each donor's residual distribution and stratum counts.]

Interferon tone is lost *before* invasion



two readouts with different confounds agree on the preinvasive loss (all padj $< 10^{-3}$)

context residual and stage enrichment — different confounds — agree on the preinvasive loss



This is the result I find most striking, and it inverts the intuition.

At the preinvasive step, two independent readouts agree. The context residual shows JAK-STAT down and IRF9 — a component of the complex that executes canonical type-one interferon transcription — stably down. And the whole-transcriptome enrichment analysis, which knows nothing about my model, shows both hallmark interferon responses down.

Those two readouts have *different* confounds. A stage contrast is vulnerable to compositional change between stages. A context residual conditions on cell state and niche stratum. When they agree across that difference, that is the strongest form of internal support I have.

Then at invasion they diverge — interferon comes back in the enrichment analysis while the residual elevates a different STAT arm.

I want to be careful here rather than claim both halves. The invasive front has documented compositional shifts, so a stage contrast is vulnerable exactly where a conditioned residual is not. The divergence localises that confound to the invasion edge. The preinvasive loss is the defensible half, and it is the half that carries the implication.

[own the divergence out loud — volunteering it is what makes the preinvasive claim credible]

And in lesions with known outcomes, that loss predicts progression

Progressive lesions: interferon lost

Regressive lesions: interferon retained

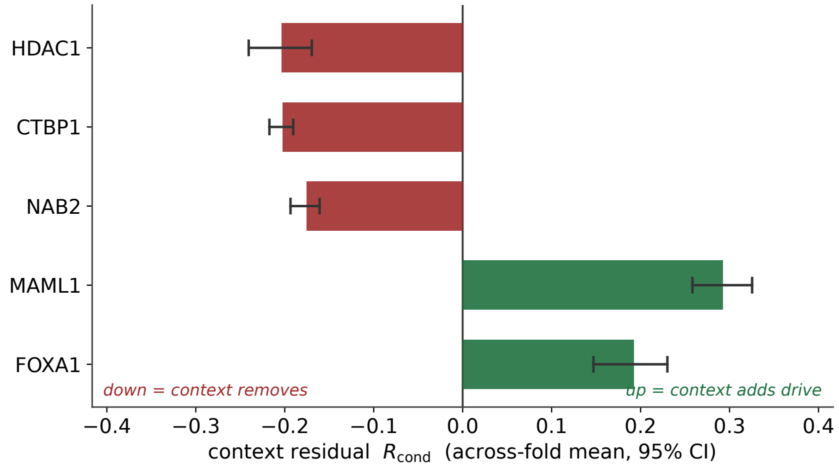
the only human premalignancy cohort profiled *and* followed to a known fate

And in lesions with known outcomes, that loss predicts progression
Progressive lesions: interferon lost
Regressive lesions: interferon retained
the only human premalignancy cohort profiled *and* followed to a known fate

And there is one cohort that can turn that into a statement about risk.
Bronchial premalignant lesions have been monitored longitudinally — profiled molecularly, and then followed until you know which ones progressed. It is, as far as I know, the only human premalignancy cohort where both halves exist.
In that cohort, the progressive and persistent lesions show decreased interferon signalling and decreased antigen presentation, with immunofluorescence confirming depletion of immune cells. The regressive lesions retain interferon tone.
Those are squamous airway precursors, not adenocarcinoma, so the comparison crosses lesion types and I want to say that plainly rather than let it slide.
But the claim it licenses is specific: loss of interferon tone in a premalignant lesion is associated with progression rather than regression. And my analysis places that loss at the preinvasive step.
There is a supporting observation from my own primary cohort: the proinflammatory macrophage niches most associated with precursor epithelium are *more* frequent in precursors and become less frequent in established cancer.
If that is right, the window in which immune-directed interception is most likely to matter closes before the transition that current staging emphasises.

[deliver as an implication, not a claim] [if pressed: Peng shows targeting inflammation in the precancerous phase

Invasion is a change of architecture, not more drive

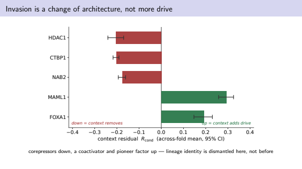


corepressors down, a coactivator and pioneer factor up — lineage identity is dismantled here, not before

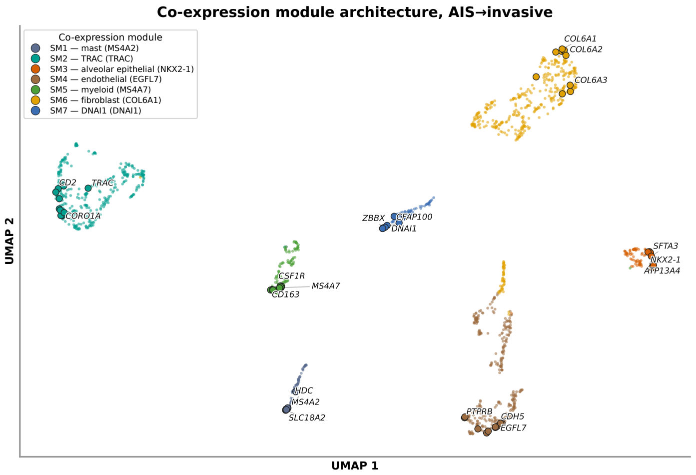
2026-07-31

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└─Invasion is a change of architecture, not more drive



Invasion is a change of architecture, not more drive

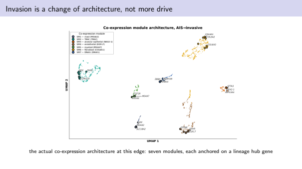


the actual co-expression architecture at this edge: seven modules, each anchored on a lineage hub gene

2026-07-31

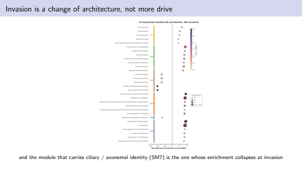
Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

└Invasion is a change of architecture, not more drive



Invasion is a change of architecture, not more drive

and the module that carries ciliary / axonemal identity (SM7) is the one whose enrichment collapses at invasion



The invasion edge is not a stronger version of the preinvasive edge. Its stable set has a different architecture, and the architecture is the finding.

Three corepressors lose context weight: HDAC1, CTBP1, and NAB2. Two factors that open chromatin gain it: MAML1, which is the Notch coactivator that recruits p300, and FOXA1, which is a pioneer factor.

MAML1 also promotes p53 acetylation after DNA damage — same axis as KAT5, preinvasively.

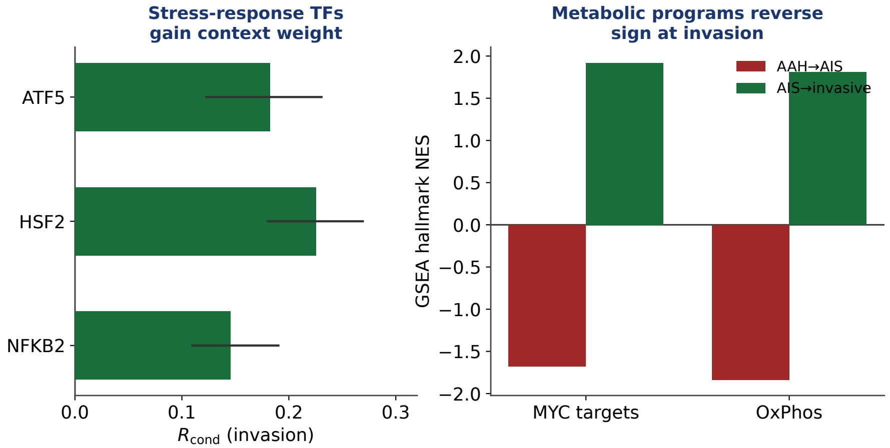
Five regulators, all operating at the level of transcriptional *coregulation* rather than sequence-specific activation, all moving the same way — toward de-repression and accessibility.

The preinvasive edge removed specific repressors from a specific target module. This edge shifts the global balance between repressive and activating coregulation. That is a different kind of change.

[advance] You can see that architecture directly. An independent co-expression analysis on this edge resolves into seven modules, each anchored on a lineage hub — fibroblast collagen, endothelial, myeloid, T-cell, and a ciliary module.

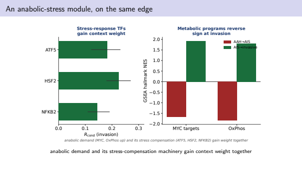
[advance] And when I score those modules for functional enrichment, the whole-transcriptome result agrees with the coregulator story. Ciliary and axonemal programs are *up* preinvasively and collapse at invasion — sensory perception of chemical stimulus is the single strongest enrichment on either edge.

An anabolic-stress module, on the same edge



anabolic demand (MYC, OxPhos up) and its stress compensation (ATF5, HSF2, NFKB2) gain weight together

anabolic demand and its stress-compensation machinery gain context weight together



There is a second module at invasion where both readouts agree.

The context residual elevates ATF5, which is an integrated-stress-response effector; HSF2, from the heat-shock family; and NFKB2, the non-canonical NF-kappa-B subunit.

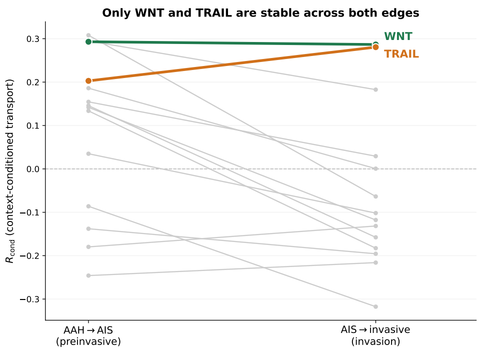
And the enrichment analysis on the same edge shows MYC targets up and oxidative phosphorylation up — both of which were *down* at the preinvasive step.

Put those together and they describe one state. High MYC output with restored oxidative metabolism is a cell operating under elevated biosynthetic load. ATF5 and heat-shock activity are exactly the compensation arms that such a load recruits. So anabolic demand and its stress-compensation machinery gain context weight together, on the edge where metabolic programs reverse direction.

There is a cross-disease parallel that supports the timing rather than the direction: proteomics of pancreatic precursors places mitochondrial remodelling prominently in high-grade lesions *before* invasion. Different trajectory, different tissue — but both put metabolic reorganisation among the earlier events.

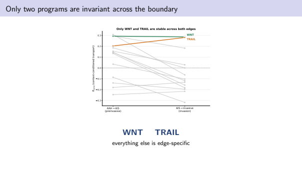
[keep this one brisk, it is a supporting module not a headline]

Only two programs are invariant across the boundary



WNT TRAIL

everything else is edge-specific



Now compare the two edges directly.

Of all the programs stable on either edge, exactly two hold both stability and sign across both: WNT and TRAIL. Everything else is edge-specific — p53, EGFR, FOS and the repressors preinvasively; VEGF, MAML1, FOXA1 and the corepressors at invasion.

That reframes WNT. On its own, WNT in a cancer context is unsurprising. But this is a specific claim: WNT context-redirection is the one regulatory effect that does *not* change while the coregulator architecture, the metabolic direction, the interferon tone, and the lineage identity all change around it.

That also makes it the more tractable interception target, because acting on an invariant effect does not require knowing which side of the invasion boundary a lesion is on — and the whole diagnostic problem is that stage does not resolve that.

I should add one caution about TRAIL. It is a death-receptor response footprint, and sustained sub-lethal death-receptor signalling in apoptosis-resistant cells is documented to produce non-apoptotic output instead of cell death. My estimand cannot distinguish those two modes, so I am not going to tell you which one this is.

[the TRAIL caveat is unprompted honesty that costs you nothing and buys a lot]

VEGF goes *down* at invasion — and that is consistent

$$\text{VEGF} \quad R_{\text{cond}} = -0.318$$

largest stable pathway at the invasion boundary

A residual is not a level.

VEGF goes *down* at invasion — and that is consistent

VEGF $R_{\text{cond}} = -0.318$

largest stable pathway at the invasion boundary

A residual is not a level.

I want to raise a result that looks wrong, before anyone raises it for me.

VEGF is the largest-magnitude stable pathway at the invasion boundary, and it is redirected *downward*. Invasion is canonically accompanied by angiogenesis. So on the face of it, the model has the sign backwards.

It does not, for two reasons.

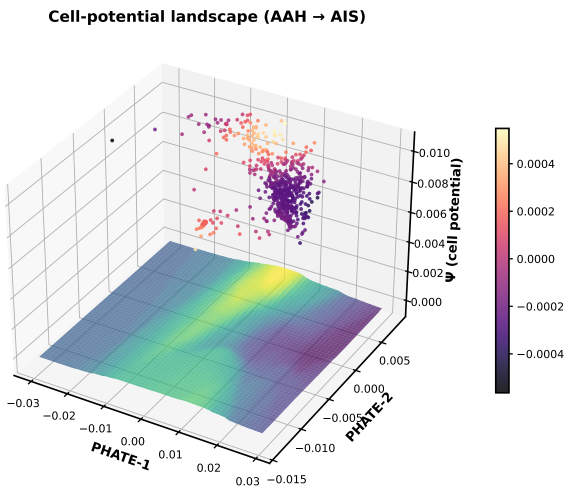
First: this is a residual, not a level. It says the surrounding tissue redirects a cell's VEGF response *relative to where its own state would have taken it*. It makes no claim about whether absolute VEGF signalling is higher or lower at invasion. A pathway can rise across a transition while its context-attributable component is negative.

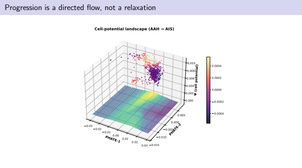
Second: the receiver here is epithelial, and the score measures VEGF *response in the cells being scored*. Angiogenic VEGF signalling is transduced predominantly by endothelium. Epithelial VEGF response and tissue-level angiogenesis are simply different quantities.

And my own data supports that separation. In the same cohort, the vascular co-expression module is enriched for migration and angiogenesis, and the stromal matrix module gains significance specifically at invasion. The tissue-level vascular program is active while the epithelial residual is negative. That is a compartment-resolved statement, and it is the kind of thing only a receiver-centred residual can make.

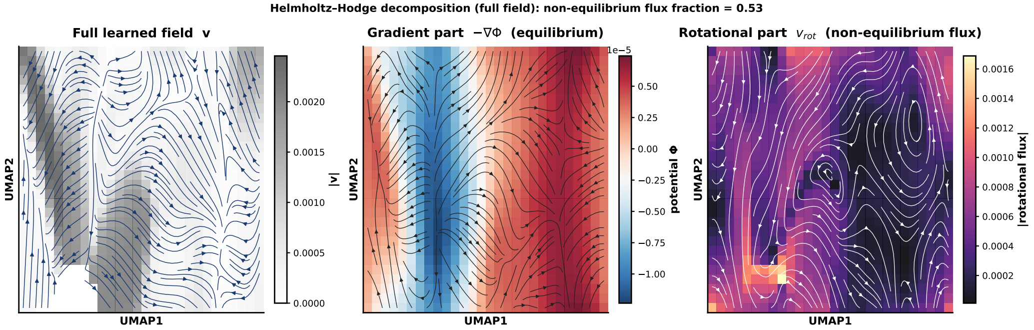
[independent support: Chen 2025 finds endothelial and stromal activity, not epithelial state, distinguishes subtypes

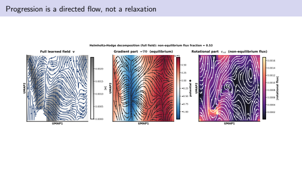
Progression is a directed flow, not a relaxation





Progression is a directed flow, not a relaxation





A geometric read of the same fitted field, because it says something the coefficients do not.
[build 1 — the landscape] First, the cell-potential landscape. The fitted field defines a scalar potential over the manifold; cells sit in it like a Waddington surface. But a potential alone would mean progression is pure downhill relaxation — and it is not.
[build 2 — Helmholtz] Decompose the field into a gradient part (equilibrium, the landscape) and a rotational part (non-equilibrium flux). On the preinvasive edge the non-equilibrium fraction is 0.53; at invasion it rises to 0.76. The invasion transition is the most strongly non-gradient — a directed, cycle-like flow, not relaxation to a fixed point. And adding context changes this fraction by at most 0.01 — the non-equilibrium character is a property of the intrinsic progression field.

[this is the systems-biology beat — "landscape AND flux," the Wang framework. Say the two numbers: 0.53 preinvasive, 0.76 invasion.] [do NOT claim thermodynamics — it is the geometry of the FITTED field, descriptive]

The gate recovers the return arm of a known circuit

$\sim$1.3 million candidate signalling chains

$\downarrow$

two programs

macrophage APOE, PSAP, A2M $\rightarrow$ LRP1, EGFR $\rightarrow$ AP-1

The gate recovers the return arm of a known circuit

$\sim$1.3 million candidate signalling chains

$\downarrow$

two programs

macrophage APOE, PSAP, A2M $\rightarrow$ LRP1, EGFR $\rightarrow$ AP-1

One more biological result, and this one connects to a circuit that has been functionally validated by someone else.

I took roughly one point three million candidate signalling chains — sender population, ligand, receptor, downstream regulator, terminal program — and gated them on the context residual. Two programs survive.

The ligands that come through are macrophage- and stromal-derived: APOE, PSAP, A2M, routed through LRP1 and EGFR onto an AP-1 program. Three of those ligands share a receptor. A filter returning noise would not reconstitute a shared-receptor ligand family from a million candidates.

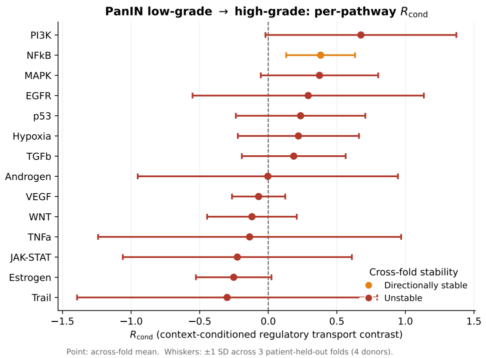
Now the connection. In mouse lung, KRAS-mutant alveolar cells secrete amphiregulin, activate EGFR on adjacent fibroblasts, drive a fibrotic program, and those fibroblasts in turn reprogram alveolar macrophages and reinforce epithelial plasticity. Blocking that axis abrogates tumour initiation entirely.

That circuit runs *from* epithelium *to* stroma. What my gate found is the return path — macrophage and stromal signals coming back onto epithelial receptors and terminating on AP-1. EGFR appears on both arms. The gate had no knowledge of that circuit.

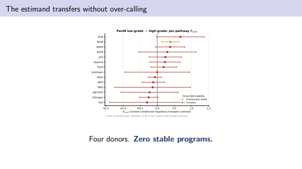
So: the epithelium builds a niche, and the niche redirects the epithelium that built it.

[frame honestly — the gate is a filter, selective by construction. The question is whether what survives is coherent. It is.] [these are nominations for organoid perturbation, not established channels]

The estimand transfers without over-calling



Four donors. **Zero stable programs.**



Hypothesis three: does this transfer?

I applied the identical estimand, unchanged, to a graded pancreatic precursor series — a biologically different epithelium, dominated by fibroblast remodelling rather than inflammatory macrophage proximity. The stability filter returns zero programs. Directionally, the biology is coherent — NF-kappa-B up, EGFR up, which is the inflammatory-to-proliferative shift the PanIN literature describes — but nothing survives the reproducibility criterion.

I want to be careful about how I describe that, because there is a tempting overstatement available and I am not going to make it.

It would be convenient to call this correct-null calibration — proof that the estimator does not manufacture findings where there is no power. But with four donors and no positive control at matched sample size, a true null and a simple absence of power are not separable. I cannot distinguish them.

So what I claim is narrower: the estimand transfers, and it declines to over-call. The experiment that would actually earn the calibration claim is to inject a synthetic effect of known magnitude at four donors and show the filter recovers it. That has not been run.

[this is a trap slide — do NOT say "correct-null calibration" out loud]

Act III

At what spatial scale does this hold?

Act III

At what spatial scale does this hold?

The falsification ladder

Rung	What it removes
Local context	nothing — reference
Shuffled context	**niche identity, distribution kept**
Donor–stage average	within-specimen variation
Self-only	context entirely
Strict-state coupling	the ecological stratification

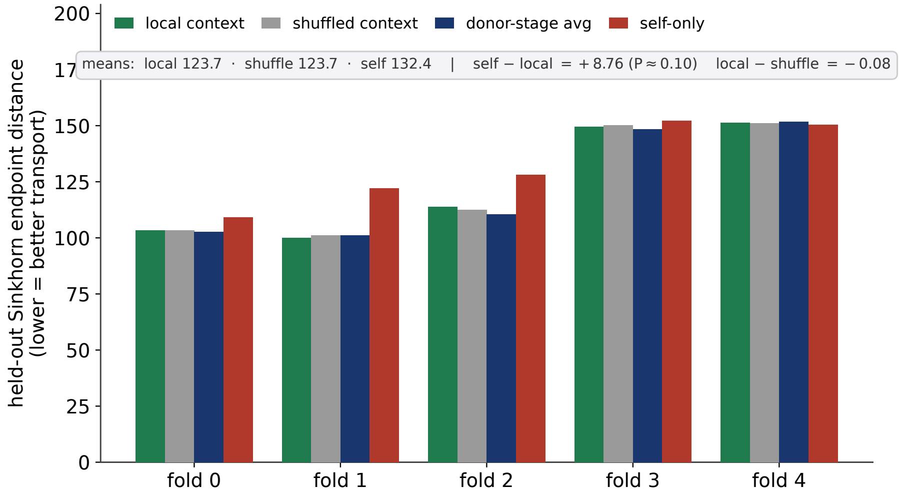
paired across five patient-held-out folds

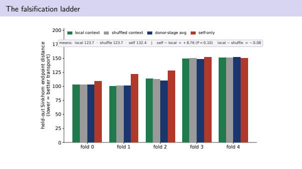
The falsification ladder

Rung	What it removes
Local context	nothing — reference
Shuffled context	**niche identity, distribution kept**
Donor–stage average	within-specimen variation
Self-only	context entirely
Strict-state coupling	the ecological stratification

paired across five patient-held-out folds

The falsification ladder





Hypothesis two is the one that required the most machinery, so here is the design.

It is a ladder of models. Each rung removes exactly one thing and everything else is held fixed, and each is scored by how well its predicted endpoints match the observed target-stage population on held-out patients.

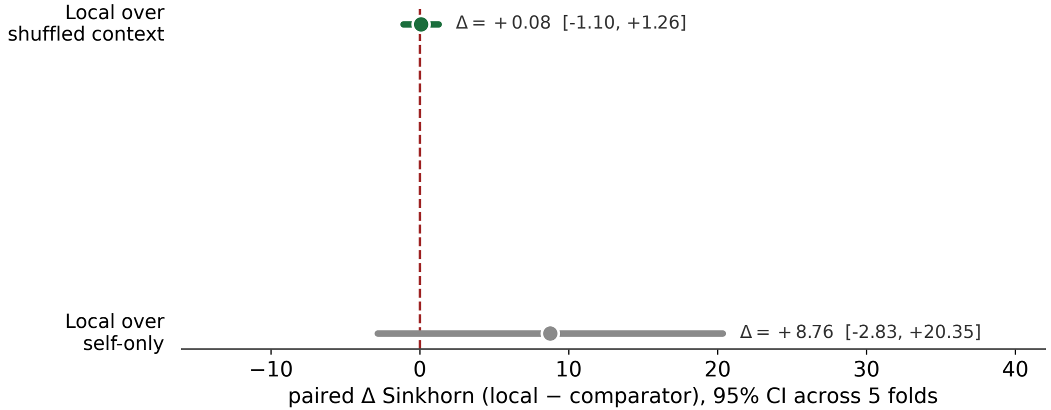
The reference is the full local-context model. Then: replace each cell's context with its donor-and-stage average, which removes within-specimen variation. Remove context entirely. Remove the ecological stratification from the coupling. Remove each half of the structured decomposition.

And the one in bold, which is the important one — shuffle the context. Permute which cell gets which neighbourhood, matched on stage and state magnitude, so the distribution of context is exactly preserved and only its *identity* is destroyed. That is the comparison that separates ecological information from added model flexibility, and it is the failure mode I flagged at the start.

Every comparison is paired within fold. Same folds, same donors, same populations — only the model changes. That is what makes the intervals on the next slide mean something.

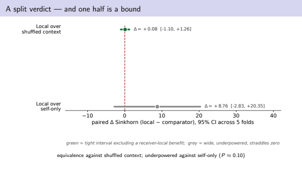
[the full ladder is twelve rungs; this is the load-bearing subset. Appendix C.10 has all of them per fold.]

A split verdict — and one half is a bound



green = tight interval excluding a receiver-local benefit; grey = wide, underpowered, straddles zero

equivalence against shuffled context; underpowered against self-only ($P \approx 0.10$)



And here is the answer, which comes in two halves that have to be kept separate.

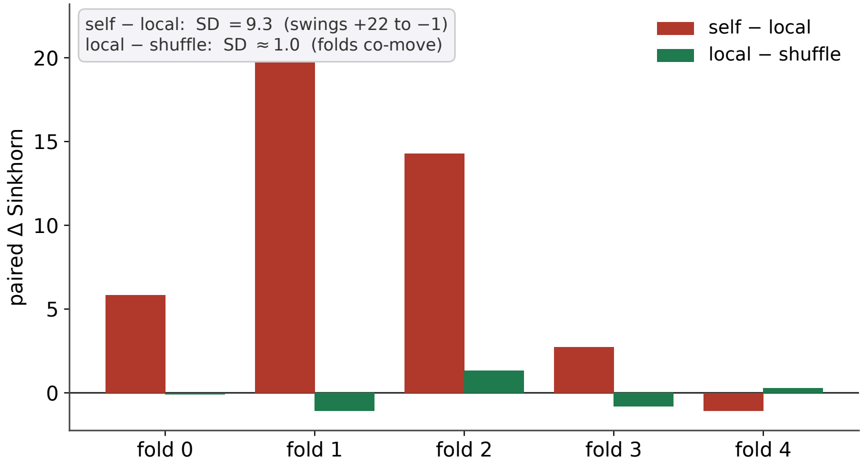
Against shuffled context, the paired difference is plus zero point zero eight, with a confidence interval from minus one point one to plus one point two six. That interval excludes any receiver-local benefit above about one point three Sinkhorn units — which is under fifteen percent of the self-only gap. That is an *equivalence* result. Not a failure to find a difference: a bound on how large any difference could be.

Against self-only, the paired difference is plus eight point seven six, but the interval runs from minus two point eight three to plus twenty point three five. Four of five folds favour local context, and the paired test gives P around zero point one zero. So that comparison is directionally consistent and underpowered. I am not claiming it.

Which means hypothesis two is not met, and the outcome is actually weaker than the pre-registered alternative branch, which required the self-only inequality to hold. It does not hold significantly. I am reporting the weaker outcome.

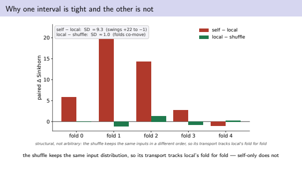
[REHEARSE THIS SLIDE VERBATIM. The trap under pressure is reverting to “improves substantially” when your own interval includes zero. Never say it.]

Why one interval is tight and the other is not



structural, not arbitrary: the shuffle keeps the same inputs in a different order, so its transport tracks local's fold for fold

the shuffle keeps the same input distribution, so its transport tracks local's fold for fold — self-only does not



The obvious question is why I trust one of those intervals and not the other, so let me answer it before it is asked.

Look at the fold-level numbers. The self-minus-local difference swings from plus twenty-two down to minus one across folds — a standard deviation of nine point three. The local-minus-shuffle difference has a standard deviation of about one.

That difference is structural, not arbitrary.

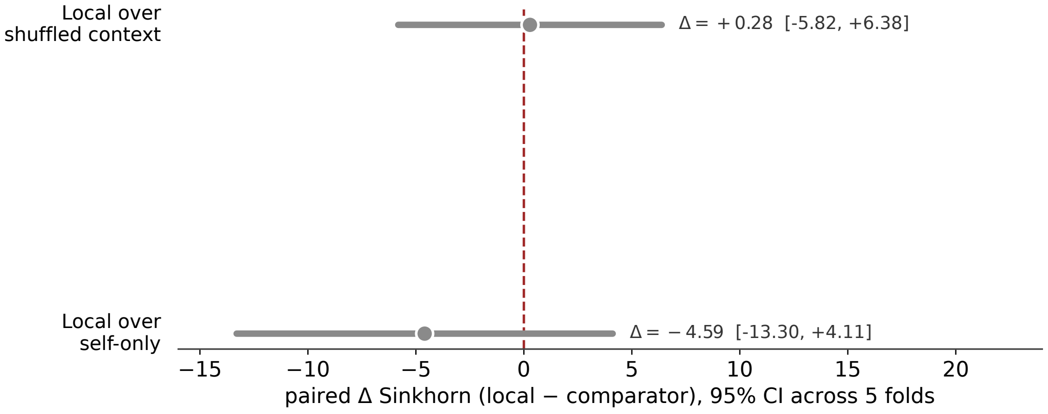
The shuffle preserves the marginal structure of the context. So the shuffled model and the local model are the same model receiving the same distribution of inputs in a different order — and their transport co-moves fold for fold. The paired difference is therefore very tightly estimated.

Self-only is a structurally different field. It is missing a whole branch. Its fold-level behaviour swings widely, and five folds simply cannot resolve an eight-point-seven-six mean gap from zero against that variance.

So one interval is trustworthy and the other is honest, and the reason is the design of the comparison rather than a choice about which result I prefer.

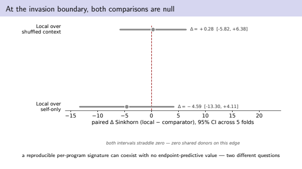
[publishing the per-fold table in Appendix C.10 is what makes the interval checkable by anyone. Say that.]

At the invasion boundary, both comparisons are null



both intervals straddle zero — zero shared donors on this edge

a reproducible per-program signature can coexist with no endpoint-predictive value — two different questions



The invasion edge behaves the same way, more starkly.

Both intervals straddle zero. Local context is statistically indistinguishable from shuffled context, and indistinguishable from no context at all. It improves over self-only in one fold out of five.

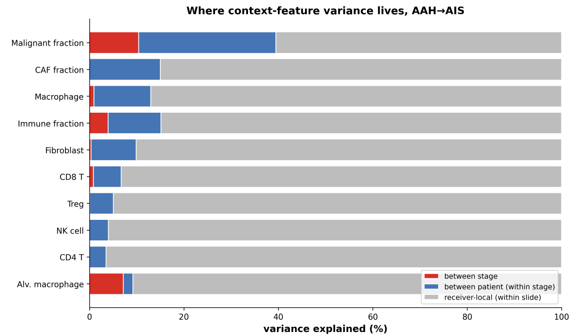
There is a structural reason. This edge has zero shared donors — no patient contributes cells to both the source and the target stage. So the within-patient contrast that a local ecological signal would need is simply not in the data.

I think the dissociation on this edge is worth naming explicitly, because it is instructive. I have a coherent, reproducible stable signature here — WNT sustained, VEGF suppressed — coexisting with no endpoint-predictive value for local context whatsoever.

Those are answers to two different questions. One is about whether per-program parameters reproduce. The other is about whether a distributional endpoint metric improves. A small reproducible effect can do the first and fail the second. This edge separates them about as sharply as they can be separated.

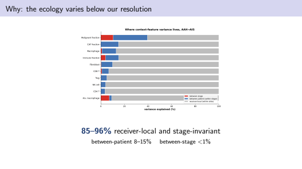
[do not present this as a second failure — present it as the same finding with the confound removed]

Why: the ecology varies below our resolution



85–96% receiver-local and stage-invariant

between-patient 8–15% between-stage <1%



So why does it fail? This is the mechanism, and I think it is the most important result in the thesis.

I decomposed the variance in the local composition features themselves — the raw ingredients of the context vector — across three levels: between stage, between patient, and receiver-to-receiver within patient.

Between-stage variance is under one percent. Between-patient is eight to fifteen. And eighty-five to ninety-six percent of the variance is receiver-local and *stage-invariant*.

Which means a cell-resolved context feature, at this resolution, is overwhelmingly carrying variation that has nothing to do with the stage contrast. It is not that there is no ecological signal. It is that the signal is embedded in a large amount of stage-orthogonal local variation, and a distributional endpoint metric cannot pull it out.

That also reconciles this with the gradient result I showed earlier, which looked like a contradiction. The gradient is a statement about parameter structure across roughly ten thousand receivers — a small systematic ordering is detectable there. The variance decomposition is a statement about how much progression-aligned signal reaches an endpoint metric. A small, reproducible effect is exactly what fails to move a distributional distance while remaining detectable per program. Same regime, two directions.

[do not rush this slide — it is the mechanism for the negative and the reason the negative is a finding]

An unrelated modality measured that scale

Immune hot and cold spots interchange over

tens of microns

Our spots are **55 µm** at **100 µm** centres

An unrelated modality measured that scale

Immune hot and cold spots interchange over

tens of microns

Our spots are **55 µm** at **100 µm** centres

And here is the part I find most persuasive, because it does not come from me.
Three-dimensional, cellular-resolution histology of more than a thousand human pancreatic precancers has mapped inflammation around individual lesions directly. What that work found is that immune hot spots and cold spots interchange over *tens of microns*.
Our spatial transcriptomic spots are fifty-five microns across, on hundred-micron centres.
So an imaging modality with no relationship to this analysis, using a completely different measurement, found the ecological variation operating at or below the resolution at which my context vector was constructed. My variance decomposition infers the same fact statistically, from transcriptomic data.
I want to be clear about what that changes. It means this is not "I could not resolve the effect." The scale at which the relevant ecology varies has been independently measured, and it sits below the assay. That converts a limitation into a measurement.

[strongest defensive move in the deck — land it slowly] [it is also the basis for a collaboration: her imaging measured the length scale, my estimator measured its consequence for inference]

Two readings, and the experiment that separates them

A. Sub-resolution

local, but below 100 μm

better assays recover it

B. Specimen-scale

a field-level property

resolution will not help

Run the identical ladder at single-cell resolution.

A predicts recovery. B predicts continued equivalence.

Two readings, and the experiment that separates them

A. Sub-resolution
local, but below 100 μm
better assays recover it

B. Specimen-scale
a field-level property
resolution will not help

Run the identical ladder at single-cell resolution.
A predicts recovery. B predicts continued equivalence.

There are two readings of that, and I cannot separate them with the data I have.

Reading A is sub-resolution. The information is genuinely receiver-local, but it varies below a hundred microns, so a higher-resolution assay would recover it.

Reading B is that the progression-relevant ecological variable is not a neighbourhood property at all — it is field-level. There is support for this: spatial mapping in donor pancreata finds the PanIN epithelium lying on a continuum with cancer while the microenvironment is drastically distinct and evolves asynchronously, and the absence of stromal reprogramming has been proposed as the reason most PanINs never progress. If that is the gating variable, it is a property of the specimen, and a receiver-local feature is measuring the wrong object however well it is resolved.

One experiment separates them: run this identical ladder on single-cell resolution spatial data. Reading A predicts recovery. Reading B predicts continued equivalence.

And I want to say what is at stake in that, because it is not small. If B holds, then microenvironment is the wrong unit of analysis for progression risk — and a substantial amount of neighbourhood-scale niche analysis has been measuring something other than what it intends to.

[that last claim is the systems-level result and the reason this work belongs in a systems journal. It is testable and consequential. Land it confidently — it is not a hedge.]

Act IV

How the hypotheses resolve

Act IV

How the hypotheses resolve

Scorecard

	Hypothesis	Verdict
H_1	Reproducible context residual	**Met**
H_2	Receiver-local information	**Not met**
	vs shuffle	bounded equivalence
	vs self-only	underpowered
H_3	Transfers across systems	**Behaviourally**

Weaker than the pre-registered alternative. I am reporting the weaker outcome.

Scorecard

	Hypothesis	Verdict
H_1	Reproducible context residual	Met
H_2	Receiver-local information	Not met
	vs shuffle	bounded equivalence
	vs self-only	underpowered
H_3	Transfers across systems	Behaviourally

Weaker than the pre-registered alternative. I am reporting the weaker outcome.

So, against the three hypotheses I stated at the beginning.

Hypothesis one is met. A reproducible, program-resolved context residual exists, and it holds across disjoint patient sets.

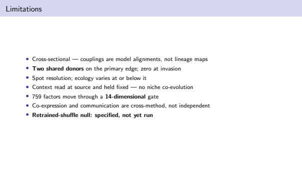
Hypothesis two is not met. And I want to give you both halves rather than the summary: against shuffled context it is a bounded equivalence, which is a quantitative result; against self-only it is underpowered, which is not.

Hypothesis three is met behaviourally — the estimand transfers and declines to over-call — but not as a demonstration of calibration, for the reasons I gave.

And one thing I want to state plainly. My pre-registration specified an alternative branch: if local context failed against shuffle but still beat self-only, that would be the specimen-ecology outcome. That branch required the self-only inequality to hold. It does not hold significantly. So the outcome I actually got is weaker than the alternative I had prepared for, and I am reporting the weaker one.

[that last paragraph is the most credibility-generating thing you will say. Do not cut it and do not rush it.] [if someone says "so H2 failed" — agree, then add that failing against shuffle with a bound excluding anything above 1.3 units is more informative than a failed test]

- Cross-sectional — couplings are model alignments, not lineage maps
- **Two shared donors** on the primary edge; zero at invasion
- Spot resolution; ecology varies at or below it
- Context read at source and held fixed — no niche co-evolution
- 759 factors move through a **14-dimensional** gate
- Co-expression and communication are cross-method, not independent
- **Retrained-shuffle null: specified, not yet run**



Seven limitations, and these are all stated in the thesis rather than discovered in the writing of this talk.
The data are cross-sectional, so the couplings are model-defined alignments, not lineage maps, and the progression coordinate carries no clock time.
Two shared donors on the primary edge and zero at invasion. This is the binding constraint on everything, and no amount of statistical care fixes it.
Receivers are decoded spots, and the ecology varies at or below that scale.
Context is read at the source stage and held fixed, so co-evolution of the niche with the cell is outside the model.
Context enters through fourteen channels, so factor-level residuals are downstream reflections of a low-dimensional gate. And the credibility diagnostic for that decomposition failed its threshold in one of five folds — reported, not hidden.
The co-expression and communication analyses read the same expression matrix, so they are cross-method consistency, not independent validation.
And the retrained-shuffle null, which is the correct calibration test for the stable program set, is specified and not yet run.

[brisk and without apology — seven real limitations stated crisply reads as command; hedging each reads as uncertainty.]

What comes next

1. Retrained-shuffle null — the outstanding confirmatory test
2. **The identical ladder at single-cell resolution**
3. Global/local decomposition, if ecology is field-scale
4. A mechanistic context channel: receptor occupancy, not a learned gate
5. Perturb APOE/PSAP/A2M $\rightarrow$ LRP1/EGFR $\rightarrow$ AP-1

Each specified precisely enough for someone else to run.

What comes next
1. Retrained-shuffle null — the outstanding confirmatory test
2. The identical ladder at single-cell resolution
3. Global/local decomposition, if ecology is field-scale
4. A mechanistic context channel: receptor occupancy, not a learned gate
5. Perturb APOE/PSAP/A2M → LRP1/EGFR → AP-1
Each specified precisely enough for someone else to run.

Five things, in the order I would do them.

First, the retrained-shuffle null. Weeks of compute, and it either confirms the twenty-nine program set or it does not. It is the single largest outstanding item.

Second — and this is the one I care most about — the identical ladder at single-cell spatial resolution. That decides between the two readings I showed you, and both outcomes are results. Recovery would mean I have measured the resolution threshold at which ecological information becomes visible. Continued equivalence would mean receiver-local niche identity does not carry progression information at any resolution.

Third, if it turns out to be field-scale, then estimate a specimen-level term explicitly rather than a receiver-local one.

Fourth, replace the learned context gate with actual receptor occupancy under known ligand-receptor structure. That makes the model a genuine universal differential equation rather than a structured approximation, and it makes the communication chains a fitted term with an estimated affinity rather than a post-hoc filter.

Fifth, perturb the axis the gate nominated, in organoid co-culture.

[point two is the one to sound most invested in — it is the paper]

A falsifiable estimand.

A screen that rejected my own first design.

Reproducible, named biology on two stage edges.

A measured bound on where it stops.

Contribution

A falsifiable estimand.

A screen that rejected my own first design.

Reproducible, named biology on two stage edges.

A measured bound on where it stops.

To summarise what I think this contributes.
A falsifiable estimand for niche effects on premalignant progression — computed out of fold, on held-out patients, read out on named biological programs rather than latent dimensions.
An identifiability screen that was capable of rejecting my own preferred design, and did.
Reproducible, signed, named biology on two stage edges, which resolves into interpretable circuitry rather than a ranked list: de-repression under retained checkpoint restraint before invasion, a coregulator switch at invasion, and interferon loss preceding the invasive transition.
And a quantitative bound on receiver-local ecological information, together with the specific experiment that decides why the recovery stops where it does.
The method works. The biology is reproducible. And the boundary is measured rather than guessed.

[closing line, out loud:] ["The result I am most confident in is the one that limits my own claim — and I can tell you exactly which experiment resolves it."] [then STOP. Do not add anything.]

Acknowledgments

Chris Bradburne

committee, collaborators, compute

Questions

Acknowledgments

Chris Bradburne

committee, collaborators, compute

Questions

Backup

Likely order: B1 null rate, B2 the 14-channel gate, B3 per-fold ladder, B4 decomposition credibility, B5 the below-chance AUC, B6 baselines, B7 metric mechanics, B8 PanIN power, B9 provenance, B10 evidential tiers, B11 spatial autocorrelation, B12 the selection screen.

B1 — Is 29/773 just the false-positive rate?

- 29/773 = 3.8%; a single 95% CI criterion has a 5% naive null rate. Fair question.
- Criterion is not one test: CI excluding zero **and** sign-consistency across five *disjoint patient* folds.
- Folds are correlated, so I do not claim an exact null rate.
- **The retrained-shuffle null is the right test and is not yet run.**
- Cheap intermediate: permute strata within donor, re-run the filter, report the null stable-count distribution.

Do not bluff. “The right test isn’t run, here is what I can bound, here is what it costs” is a strong answer. Improvising a null rate is not.

B2 — The context channel is 14-dimensional

- Context enters only via $\mathbb{R}^{35} \to \mathbb{R}^{14} \to \mathbb{R}^{773}$.
- So coordination among transcription factors is partly *expected*, not evidential.
- What the bottleneck does *not* determine is which factors load or with what sign. A rank-restricted map does not force four repressors to oppose their target. **Sign structure is the informative part.**
- Pathway-level claims sit directly in the gated block and are strongest.

B2 — The context channel is 14-dimensional

- Context enters only via $\mathbb{R}^{35} \to \mathbb{R}^{14} \to \mathbb{R}^{773}$.
- So coordination among transcription factors is partly *expected*, not evidential.
- What the bottleneck does *not* determine is which factors load or with what sign. A rank-restricted map does not force four repressors to oppose their target. **Sign structure is the informative part.**
- Pathway-level claims sit directly in the gated block and are strongest.

If you get this question you are being examined by someone who read the field equation carefully. Answering it precisely is worth more than any result on the main slides.

	F0	F1	F2	F3	F4	mean
Local context	103.32	100.12	113.93	149.47	151.42	123.65
Self-only	109.14	122.16	128.19	152.20	150.36	132.41
self − local	+5.82	+22.04	+14.26	+2.73	−1.06	+8.76

SD 9.33, SEM 4.17, $t = 2.10$, $P \approx 0.10$. Against shuffle, SD ≈ 1.0.

B3 — Per-fold ladder, preinvasive edge

	F0	F1	F2	F3	F4	mean
Local context	103.32	100.12	113.93	149.47	151.42	123.65
Self-only	109.14	122.16	128.19	152.20	150.36	132.41
self − local	+5.82	+22.04	+14.26	+2.73	−1.06	+8.76

SD 9.33, SEM 4.17, $t = 2.10$, $P \approx 0.10$. Against shuffle, SD ≈ 1.0.

B4 — Decomposition credibility per fold

	F0	F1	F2	F3	F4	mean
D_ρ	0.949	0.963	1.008	0.984	0.922	0.965

- $D_\rho \leq 1$ means structured terms carry at least as much velocity as the neural residual — the interpretability precondition.
- **Fold 2 fails it.** Reported, not hidden.
- Which is why factor-level claims are stated below pathway-level ones.

B5 — Why is the held-out AUC *below* chance?

- Donor-held-out AUC for the self-only stage classifier: **0.419**, about 1.6 SE below 0.5.
- Not a bug — the signature of a **donor-entangled representation**. Cross-donor generalisation is anti-correlated, not absent.
- **Within-donor AUC is 0.833**, above chance in every shared donor.
- The representation is stage-discriminative; cross-patient transfer is what fails. Which is exactly why every estimate is patient-held-out.

B5 — Why is the held-out AUC *below* chance?
- Donor-held-out AUC for the self-only stage classifier: **0.419**, about 1.6 SE below 0.5.
- Not a bug — the signature of a **donor-entangled representation**. Cross-donor generalisation is anti-correlated, not absent.
- **Within-donor AUC is 0.833**, above chance in every shared donor.
- The representation is stage-discriminative; cross-patient transfer is what fails. Which is exactly why every estimate is patient-held-out.

Lead with “that is the confound signature,” not “it is at chance.”

B6 — Baselines and ablations

- Linear endpoint baselines: 210.83 and 251.86 vs 123.65 fitted.
- Single-branch fields collapse: structured-only 2621.22, residual-only 654.81.
- Monolithic network fits *better* (121.59) — the price of an interpretable, program-resolved field. Stated, not omitted.
- Strict-state fits best (119.17) because the metric is scored in the space that coupling optimises. Near-circular, hence rejected on Tier 2.

B6 — Baselines and ablations
- Linear endpoint baselines: 210.83 and 251.86 vs 123.65 fitted.
- Single-branch fields collapse: structured-only 2621.22, residual-only 654.81.
- Monolithic network fits *better* (121.59) — the price of an interpretable, program-resolved field. Stated, not omitted.
- Strict-state fits best (119.17) because the metric is scored in the space that coupling optimises. Near-circular, hence rejected on Tier 2.

B7 — Endpoint metric mechanics

- Conditional metric: Sinkhorn cost within held-out strata (≥ 5 per side), source-mass weighted, $\varepsilon = 0.05$.
- Primary because it evaluates the same conditioning that *defines* the estimand.
- Comparisons paired within fold, so estimator bias common across rungs cancels.
- An orthogonal metric — donor-held-out stage classification from predicted endpoints — would harden the negative. Named as future work.

B7 — Endpoint metric mechanics

- Conditional metric: Sinkhorn cost within held-out strata (≥ 5 per side), source-mass weighted, $\varepsilon = 0.05$.
- Primary because it evaluates the same conditioning that *defines* the estimand.
- Comparisons paired within fold, so estimator bias common across rungs cancels.
- An orthogonal metric — donor-held-out stage classification from predicted endpoints — would harden the negative. Named as future work.

B8 — PanIN power

- Four donors, $\sim$100 graded receivers, two strata, three folds.
- Zero stable; 188 directionally stable.
- **True null and absence of power are not separable** without a positive control at matched n.
- The experiment that would earn the claim: inject a synthetic effect of known magnitude at $n = 4$ and show recovery.

B8 — PanIN power

- Four donors, $\sim$100 graded receivers, two strata, three folds.
- Zero stable; 188 directionally stable.
- **True null and absence of power are not separable** without a positive control at matched n.
- The experiment that would earn the claim: inject a synthetic effect of known magnitude at $n = 4$ and show recovery.

B9 — Provenance

- Every inline number traces to a versioned artifact; audit trail in the Ch3 header and Appendix C.15.
- Folds, hyperparameters, seeds: Appendix C.5–C.7.
- Per-fold ladders, 773-program stability atlas, transport diagnostics: Appendix C.8–C.11.
- Seeded per fold; training donors only; code and environment archived.

Any number on any slide can be traced live.

B9 — Provenance
- Every inline number traces to a versioned artifact; audit trail in the Ch3 header and Appendix C.15.
- Folds, hyperparameters, seeds: Appendix C.5–C.7.
- Per-fold ladders, 773-program stability atlas, transport diagnostics: Appendix C.8–C.11.
- Seeded per fold; training donors only; code and environment archived.

Any number on any slide can be traced live.

B10 — Secondary analyses and their evidential status

Tier	Analyses
Model-independent	stage DE, gene-set enrichment, stage discrimination
Residual-associated	ecological localisation, spatial autocorrelation
Residual-guided	SCENIC / hdWGCNA, gated communication

- Only the first tier is independent of the estimand.
- SCENIC overlap is **not enriched** beyond hypergeometric expectation (0.95–1.14$\times$, $P = 0.35$–0.63). Reported as such.

B10 — Secondary analyses and their evidential status

Tier	Analyses
Model-independent	stage DE, gene-set enrichment, stage discrimination
Residual-associated	ecological localisation, spatial autocorrelation
Residual-guided	SCENIC / hdWGCNA, gated communication

- Only the first tier is independent of the estimand.
- SCENIC overlap is **not enriched** beyond hypergeometric expectation (0.95–1.14$\times$, $P = 0.35$–0.63). Reported as such.

Claiming independence for the third tier is the easiest way to lose credibility you have already earned.

B11 — The ecology is spatially structured

- Moran's I, $k = 15$ per section, 55 sections, 787,816 spots. All 330 section-by-feature tests significant.
- Malignant clustering rises monotonically: $0.234 \to 0.458 \to 0.701$; Spearman $\rho = 0.673$.
- Getis–Ord hot-spot fraction nearly doubles.
- But macrophage autocorrelation is high and **stage-flat** ($0.443 \to 0.499 \to 0.467$).

Spatially organised within every specimen, only weakly graded by stage — the regime where local niche is real but hard to separate.

B11 — The ecology is spatially structured

- Moran's I, $k = 15$ per section, 55 sections, 787,816 spots. All 330 section-by-feature tests significant.
- Malignant clustering rises monotonically: $0.234 \to 0.458 \to 0.701$; Spearman $\rho = 0.673$.
- Getis–Ord hot-spot fraction nearly doubles.
- But macrophage autocorrelation is high and **stage-flat** ($0.443 \to 0.499 \to 0.467$).

Spatially organised within every specimen, only weakly graded by stage — the regime where local niche is real but hard to separate.

B12 — The estimand selection screen

- Frozen before real data: recovery cosine ≥ 0.80, seed agreement ≥ 0.90, null gain < 0.10, non-identifiable correlation < 0.50.
- Five regimes including a confounded null and an alignment sweep.
- Conditional-stratified passed both tiers. Strict-state passed tier 1, failed tier 2. Progression-geometry failed both.
- Oracle pairing was the ceiling and is unavailable to any real analysis.

Two of three candidates rejected, including the one I started with.

B12 — The estimand selection screen

- Frozen before real data: recovery cosine ≥ 0.80, seed agreement ≥ 0.90, null gain < 0.10, non-identifiable correlation < 0.50.
- Five regimes including a confounded null and an alignment sweep.
- Conditional-stratified passed both tiers. Strict-state passed tier 1, failed tier 2. Progression-geometry failed both.
- Oracle pairing was the ceiling and is unavailable to any real analysis.

Two of three candidates rejected, including the one I started with.